IEM UTM STUDENT SECTION

UTM
UNIVERSITI TEKNOLOGI MALAYSIA

IEM Connect

Sprint 1 Presentation

Presented by: Group GGEZ

Date: 10 November 2025

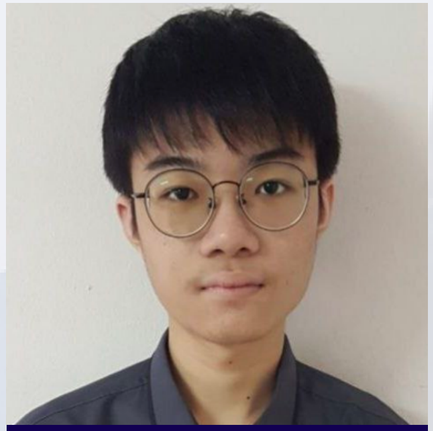
Play

Our Team

Chuah Chun Yi

Scrum Master

Lead Coordination & Stakeholder Communication



Chong Jun Hong

Developer 1

Backend Development

Muhammad Ariff bin Baharuddin

Developer 2

Frontend Development

Huzsa Syahyzal Haiqal bin Hussain

Developer 3

Frontend Development

Hrishi Vengalsetti

Developer 4

Backend Development

Sprint Overview

Sprint Goal

- To develop the core features of the IEM Connect System that allow users to register, login, verify their login with a Temporary Access Code and enable admin to review and verify member registrations.

Sprint Duration

- 2 Weeks (Design + Development + Testing)

MVP Objective

- Deliver a functional registration and login module that apply 2 Factors Authentication integrated with the membership database.
- Deliver a working admin module where admin able to review and verify member registrations.

Sprint Backlog Summary

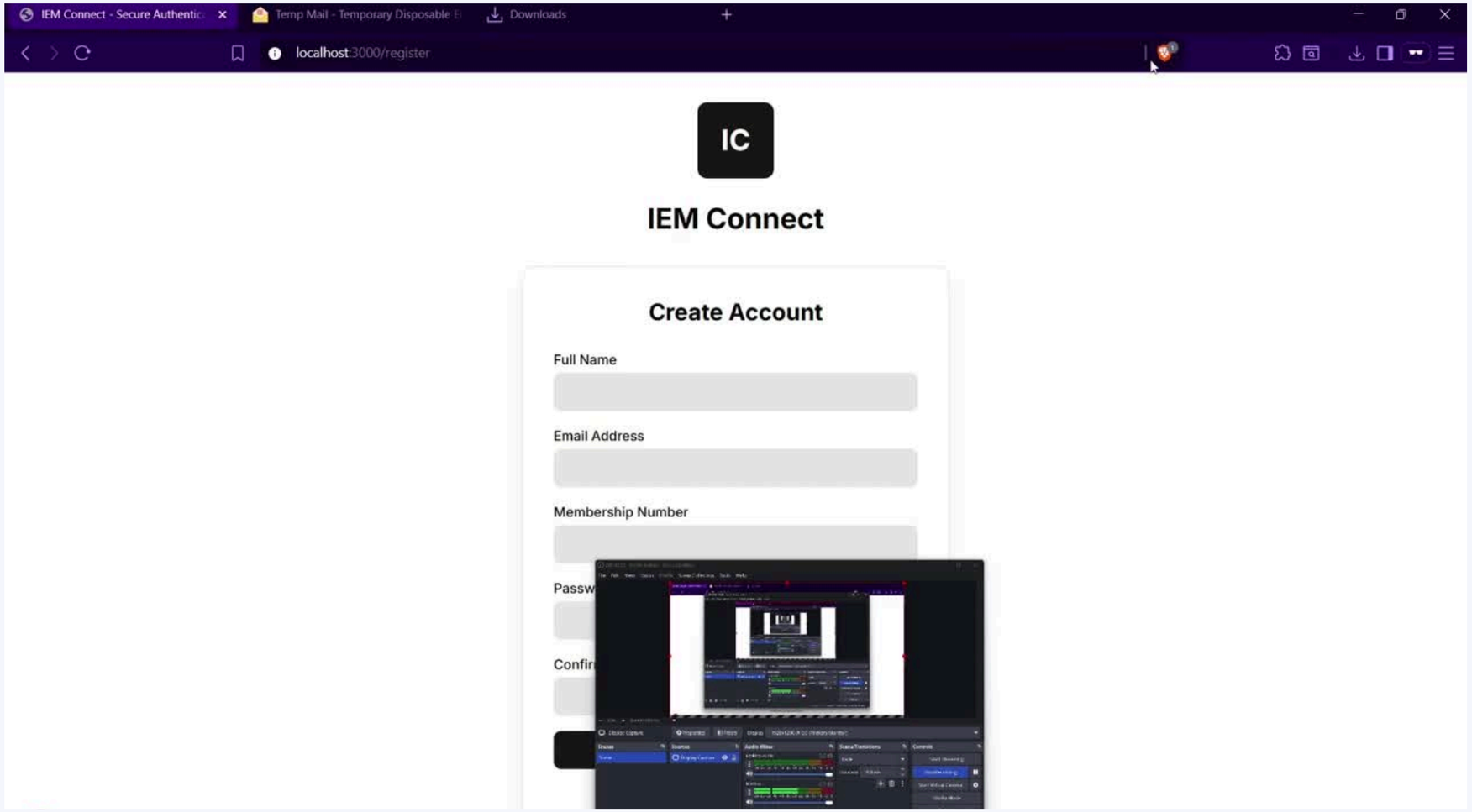
Main User Stories Implemented:

1. As a new user, I want to register and login so that I can access IEM Connect securely.
2. As a member, I want to verify my login with a Temporary Access Code so that my account is secured from unauthorized access.
3. As an admin, I want to review member registrations and verify membership numbers so that only verified IEM members can access IEM Connect.

Deliverables:

- User Registration and Login
- 2 Factor Authentication
- Admin Review and Verify

MVP Demonstration



Client Feedback

Key Comments Received:

1. System meets expected security and functionality standards.
2. Suggested UI enhancements and dashboard analytics for the next sprint.
3. Suggested using institution email validation for the auto-approval process.
4. Requested integration of member verification and event dashboard in the next sprint.

Action Items for Next Sprint:

- Set up pre-defined test scenarios for Sprint 2.
- Implement notification component for registration feedback.

Code Hightlights

Core Components

controller/authController.js:

- handles registration and input validation
- handles user login
- handles 2FA Verification

controller/adminController.js:

- Admin can review and verify users’ registration

models/User.js:

- Defines the users table with all authentication fields

Example Logic

1 Registration Logic

```
export const register = async (req,res)=>{
const { name,email,password } = req.body;
const hashed = await
bcrypt.hash(password,10);
const user = await User.create({
name,email,password_hash:hashed });
res.json({ message:"User registered",
id:user.id });
};
```

Passwords are hashed using bcrypt for security before saving to the database.

2 Login Logic

```
export const login = async (req,res)=>{
const user = await User.findOne({ where:
{email:req.body.email} });
const valid = await
bcrypt.compare(req.body.password,user.password_hash);
if(!valid) return res.status(401).json({msg:"Invalid
password"});

const otp = generateOTP();
const hash = await bcrypt.hash(otp,10);
await user.update({ two_fa_code_hash:hash,
two_fa_code_expiry: Date.now()+300000 });
console.log("OTP:", otp);
res.json({ msg:"2FA code sent" });
};
```

Verify credentials, then generate and save a hashed OTP for two-factor authentication.

Example Logic

3 Admin Review & Verify

```
export const getPendingUsers = async (req,res)=>{
const users = await User.findAll({ where:{
is_verified:false } });
res.json(users);
};
export const approveUser = async (req,res)=>{
await User.update({ is_verified:true },{ where:{
id:req.params.id } });
res.json({ msg:"User approved by admin" });
};
```

Admin can view all pending users and approve their verification status.

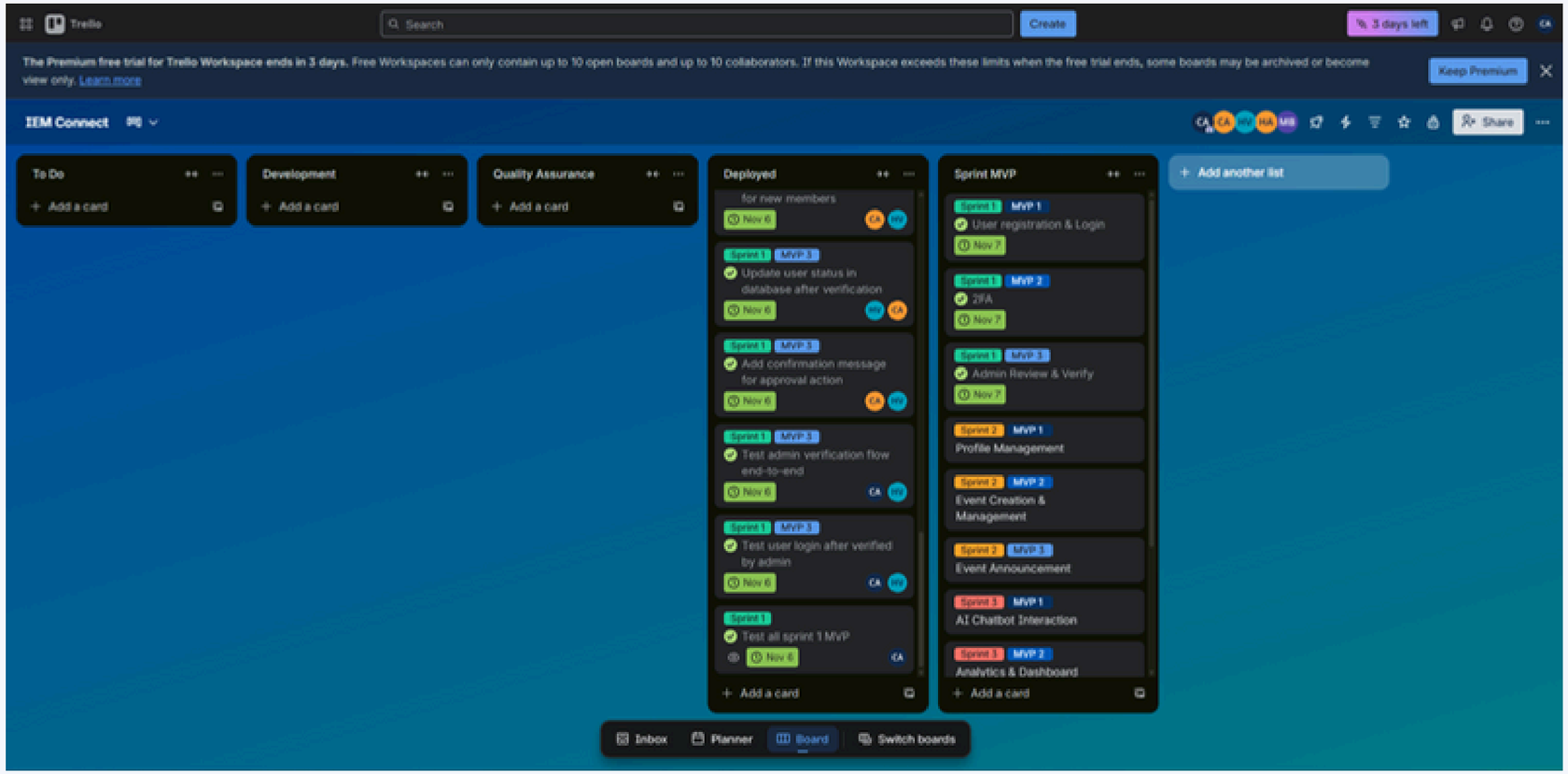
4 2FA Verification

```
export const verify2FA = async (req,res)=>{
const { email,otp } = req.body;
const user = await User.findOne({ where:{email} });
const match = await
bcrypt.compare(otp,user.two_fa_code_hash);
if(!match || Date.now()>user.two_fa_code_expiry)
return res.status(400).json({msg:"OTP invalid or
expired"});
await user.update({ is_verified:true,
two_fa_code_hash:null });
res.json({ msg:"2FA verified" });
};
```

Confirms OTP validity and marks the user as verified, removing OTP data from the database.

Project Performance Metrics

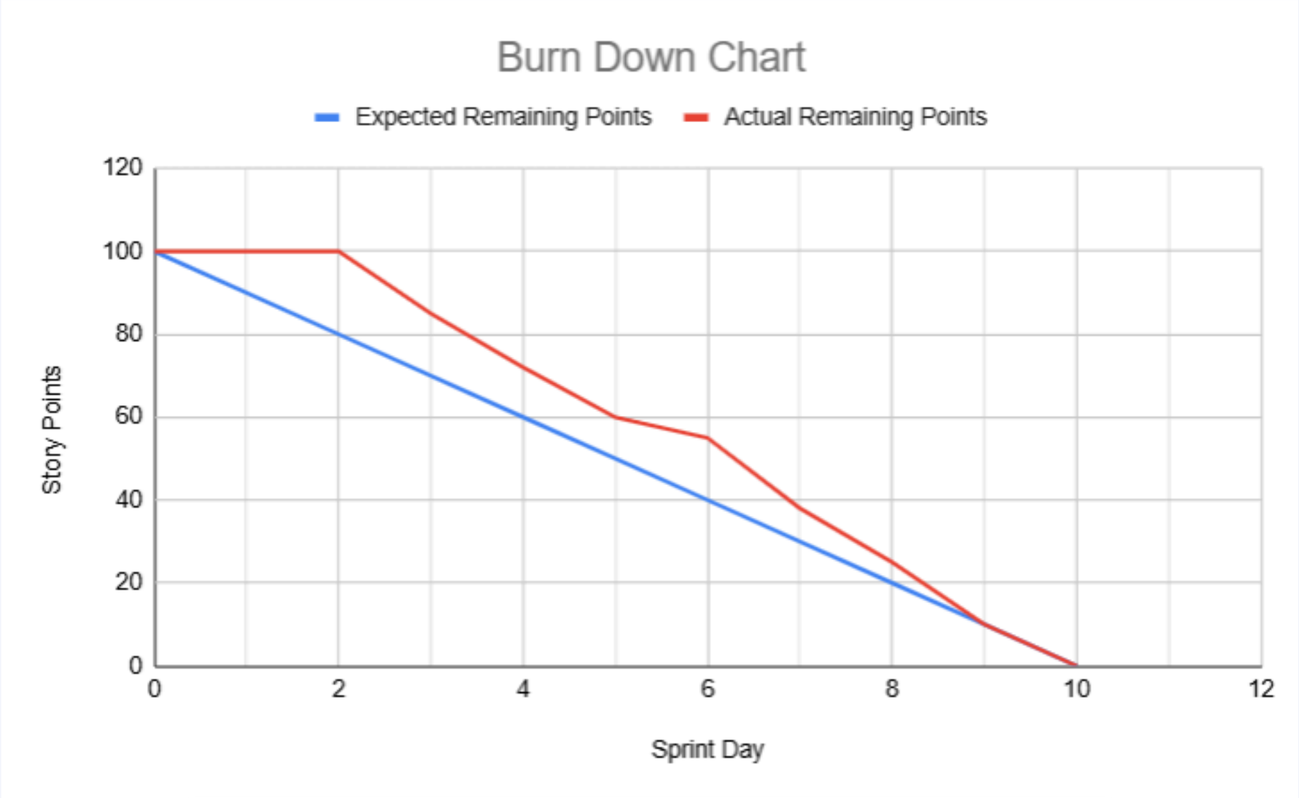
Kanban Board



Sprint 1
Total tasks: 15
Completed: 15
In progress: 0

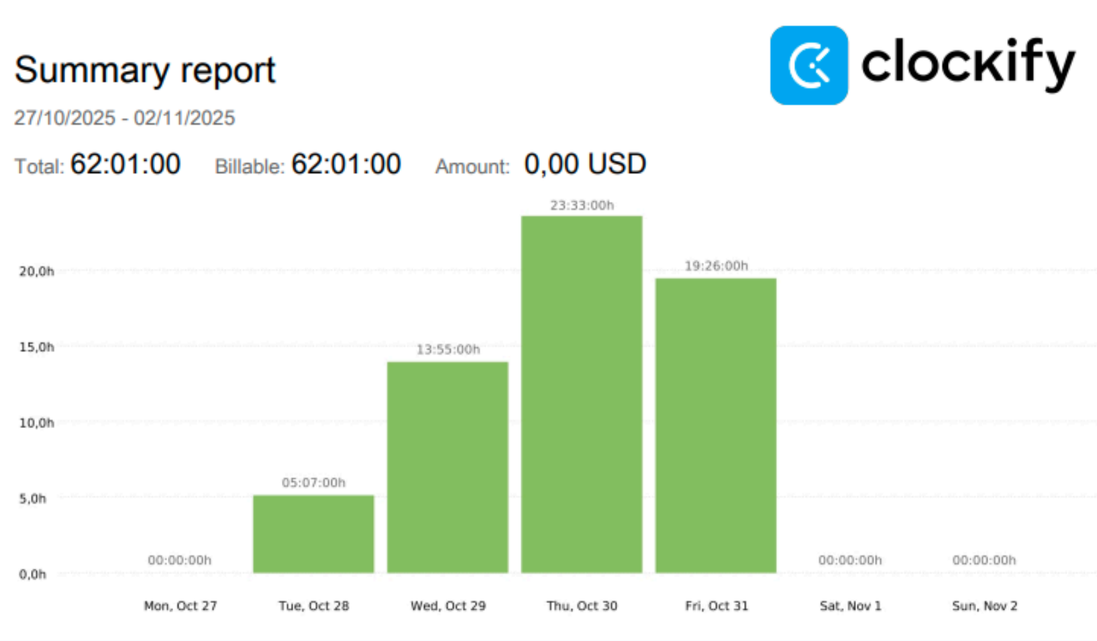
Project
Total tasks: 11
Completed: 3
In progress: 8

Burndown Chart



Sprint completed on time with all tasks achieved

Clockify Summary



Total hours logged: 128.67 hours
Average per member: 25.73 hours

Next Sprint Focus

Sprint 2 Objectives:

- Implement profile management features for members
- Develop event creation and management functionalities for admin
- Integrate event announcement tools within admin functions to enable member notifications.

Target Outcome:

- Deliver an improved MVP with implemented members' profile management functions and full admin's event management functionalities.

Conclusion

Summary:

- Sprint 1 successfully delivered a working MVP with working member registrations plus login system integrated with 2 Factor-Authentication and the admin review and verification functionalities.

Next Steps:

- Apply client feedback for Sprint 2 improvement
- Improve and polish the appearance of user interface
- Prepare system testing scripts for new modules.

Thank you

Contact Us

Chuah Chun Yi

Scrum Master of Group GGEZ

+60 1157678402

iemconnect1@gmail.com

iemconnect.com.my (Coming soon)